

# Companion to Bird & de Moor, Algebra of Programming

## §2. Functions and Categories

### §2.1. Categories

## §2.2. Functors

## §2.3. Natural transformations

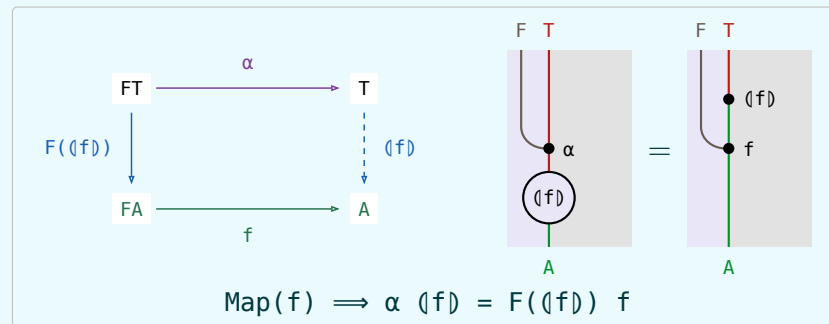
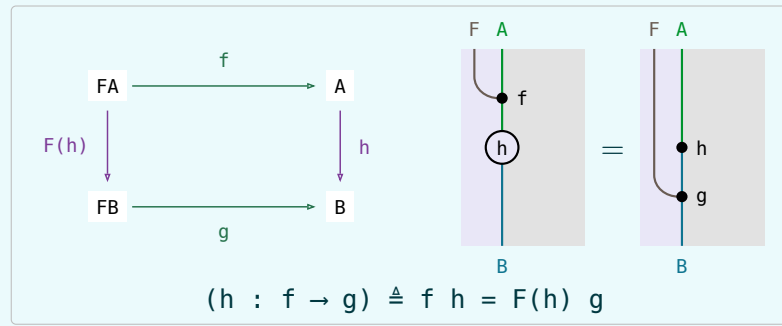
## §2.4. Constructing datatypes

## §2.5. Products and coproducts

## §2.6. Initial algebras

(2.6a)

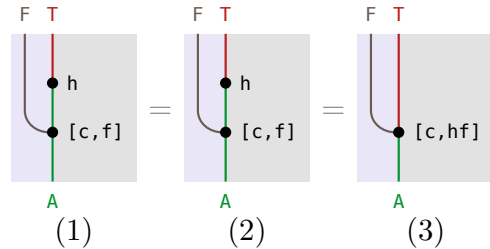
The **initial algebra**  $\alpha : F(T) \rightarrow T$  has exactly one  $F$ -homomorphism  $\langle f \rangle : T \rightarrow A$  to every  $F$ -algebra  $f : F(A) \rightarrow A$ .



$$\alpha h = F(h) [c, f] \iff \text{zero } h = c \wedge \text{succ } h = h f \quad (2.6b)$$

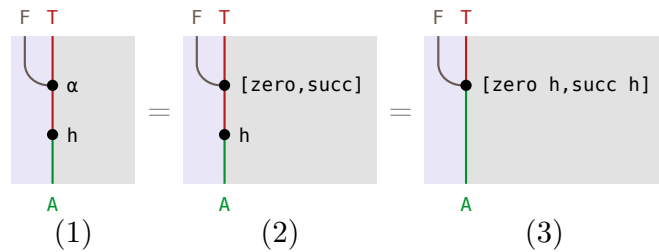
$h$  is a homomorphism from  $\alpha = [\text{zero}, \text{succ}]$  to  $[c, f]$  exactly when  $h$  sends **zero** to **c** and **succ** then  $h$  equals  $h$  then  $f$

the right side

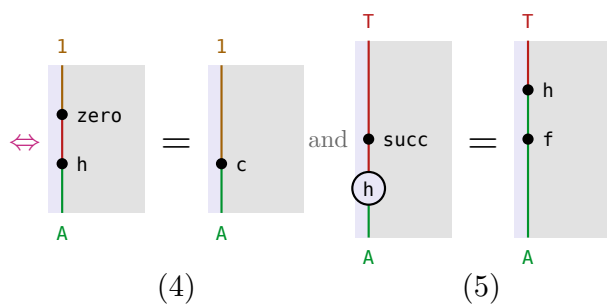


|   |                    |                   |
|---|--------------------|-------------------|
| 1 | $F(h) [c, f]$      | definition of $F$ |
| 2 | $= (1 + h) [c, f]$ | coproduct         |
| 3 | $= [c, h f]$       |                   |

the left side



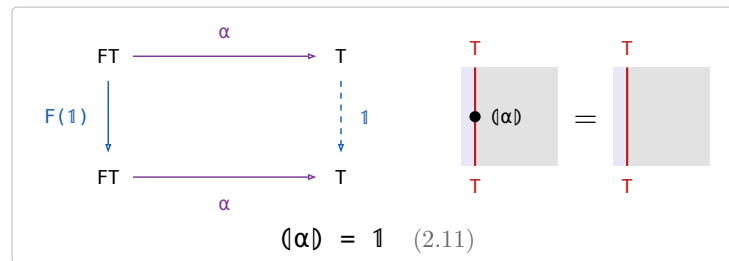
|   |                                      |                                             |
|---|--------------------------------------|---------------------------------------------|
| 1 | $\alpha h$                           | since $\alpha = [\text{zero}, \text{succ}]$ |
| 2 | $= [\text{zero}, \text{succ}] h$     | coproduct                                   |
| 3 | $= [\text{zero } h, \text{succ } h]$ | cancellation, zero arm                      |



|   |                                                                                    |                        |
|---|------------------------------------------------------------------------------------|------------------------|
| 4 | $\iff [\text{zero } h, \text{succ } h] = [c, h f] \Rightarrow \text{zero } h = c$  | cancellation, succ arm |
| 5 | and $[\text{zero } h, \text{succ } h] = [c, h f] \Rightarrow \text{succ } h = h f$ |                        |

## §2.6.1. Reflection

(2.6.1a)

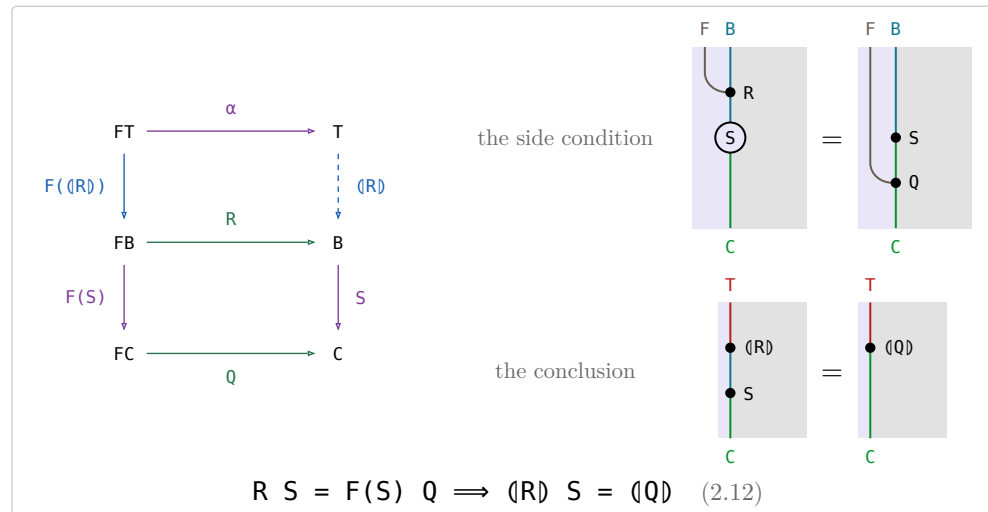


Taking a value apart with  $\alpha$  and putting it straight back is doing nothing.

**§2.6.2.  $R S = F(S) Q \implies (R) S = (Q)$  — folding with  $R$ , then applying  $S$ , is folding with  $Q$**

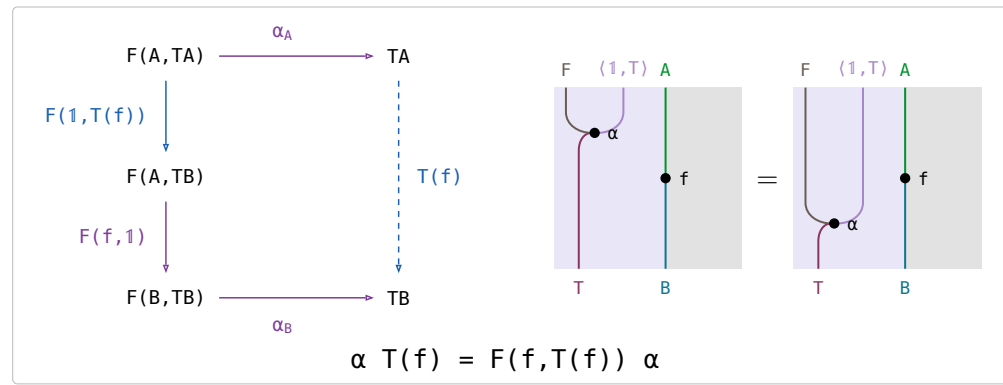
When  $S : B \rightarrow C$  is an  $F$ -homomorphism from  $R : F(B) \rightarrow B$  to  $Q : F(C) \rightarrow C$ , folding with  $R$  and then applying  $S$  is folding with  $Q$ .

(2.6.2a)

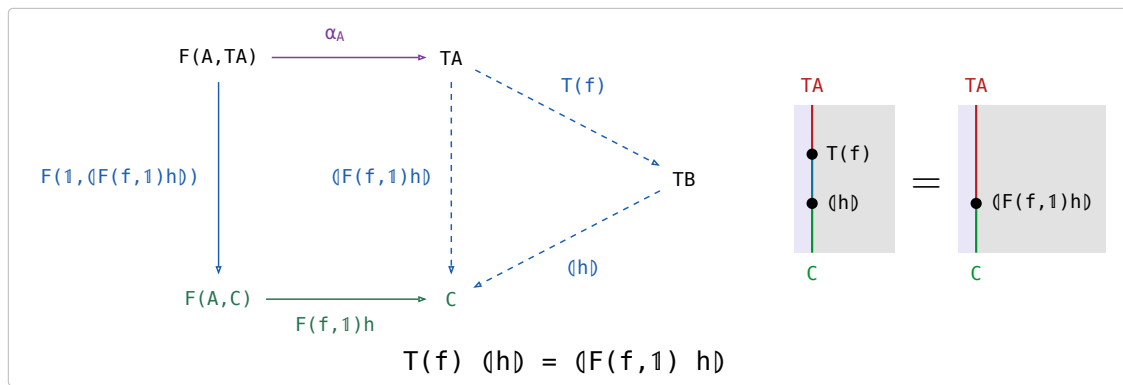


## §2.7. Type functors

$T(R) \triangleq \langle F(R, 1) \ \alpha \rangle$  — the type functor  $T$  acts on a map  $f : A \rightarrow B$  through the initial algebras  $\alpha_A : F(A, TA) \rightarrow TA$ .



- $F : \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$  is a bifunctor and a wire is a unary functor, so the two arguments are packed first:  $\langle 1, T \rangle : \mathcal{A} \rightarrow \mathcal{A} \times \mathcal{A}$  sends  $A$  to  $(A, TA)$ , and  $F(\langle 1, T \rangle(A))$  is  $F(A, TA)$ .
- The picture is three wires —  $F$ ,  $\langle 1, T \rangle$ , and the object — and the region between the first two is  $\mathcal{A} \times \mathcal{A}$ .
- $\alpha$  is then an ordinary natural transformation  $F \circ \langle 1, T \rangle \Rightarrow T$ : its bead eats the  $F$  and  $\langle 1, T \rangle$  wires, and the  $T$  wire is born under it.
- $F(f, T(f))$  costs no notation. It is the bead  $f$  on the object wire with  $\langle 1, T \rangle$  and  $F$  running past:  $\langle 1, T \rangle$  is what turns  $f$  into the pair  $(f, T(f))$ , and  $F$  is what applies it.
- The law is the naturality of  $\alpha$ , which is exactly the freedom to slide that  $f$  bead past it.



|   | name                   | law                                     | what it says                                                                                                                                                                                                |
|---|------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | functor                | $T(1) = 1$ and $T(R) \ T(S) = T(R \ S)$ | Acting by the identity changes nothing, and two actions in a row are one action.                                                                                                                            |
| 2 | type functor fusion    | $T(f) \ (h) = (F(f, 1) \ h)$            | A relator action followed by a fold is a single fold — the intermediate structure is never built. The side condition holds because $F$ is a bifunctor — $F(R, 1)F(1, (Q)) = F(R, (Q)) = F(1, (Q))F(R, 1)$ . |
| 3 | naturality of $\alpha$ | $\alpha \ T(f) = F(f, T(f)) \ \alpha$   | Building and then mapping is the same as mapping the parts and then building, so $\alpha$ is natural from $G(R) = F(R, T(R))$ to $T$ .                                                                      |